laplace transforms, systems of linear equations

For this section it would be much simpiler if we jumped into doing practice problems instead of "theroy" or explinations

Solve using the system of linear DE by using Laplace transform

$$\begin{cases} \dfrac{dx}{dt} = 2y + e^t \\ \dfrac{dy}{dt} = 8x - t \end{cases} \qquad x(0)=1, \quad y(0)=1$$

you first want to rewrite the system of equation like this:

$$\begin{cases} x' - 2y = e^t \\ y' - 8x = -t \end{cases}$$

- after you rewrite you then need to apply the laplace transform to it

$$\begin{cases} \mathcal{L}\{x'\} - 2\mathcal{L}\{y\} = \mathcal{L}\{e^t\} \\ \mathcal{L}\{y'\} - 8\mathcal{L}\{x\} = -\mathcal{L}\{t\} \end{cases} \Longrightarrow \begin{cases} sx(s) - x(0) - 2y(s) = \dfrac{1}{s-1} \\ sy(s) - y(0) - 8x(s) = -\dfrac{1}{s^2} \end{cases}$$

then plug in the IVP into the system of equation and then elimate all x's or elimate all y's

$$\begin{cases} s x(s) - x(0) - 2y(s) = \dfrac{1}{s-1} \\ s y(s) - y(0) - 8x(s) = -\dfrac{1}{s^2} \end{cases} \Longrightarrow \begin{cases} s x(s) - 2y(s) = \dfrac{1}{s-1} + 1 \\ s y(s) - 8x(s) = -\dfrac{1}{s^2} + 1 \end{cases}$$

if we multiply the first equation by 8 and the second equation by s we can cancel the x terms

$$\begin{cases} 8\left(s x(s) - 2y(s) = \dfrac{1}{s-1} + 1\right) \\ s\left(s y(s) - 8x(s) = -\dfrac{1}{s^2} + 1\right) \end{cases} \Longrightarrow \begin{cases} \cancel{8s x(s)} - 16y(s) = \dfrac{8}{s-1} + 8 \\ s^2 y(s) - \cancel{8s x(s)} = -\dfrac{1}{s} + s \end{cases}$$

$$s^2 y(s) - 16y(s) = \frac{8}{s-1} + 8 - \frac{1}{s} + s$$

$$y(s)\,[s^2 - 16] = \frac{8}{s-1} + 8 - \frac{1}{s} + s$$

$$y(s) = \frac{8}{(s-1)(s+4)(s-4)} + \frac{8}{(s+4)(s-4)} - \frac{1}{s(s+4)(s-4)} + \frac{s}{(s+4)(s-4)}$$

• we find the LCD: $(s-1)(s+4)(s-4)(s)$ and we combine them all togther to get;

$$\frac{s^3 + 7s^2 - s - 1}{s(s-1)(s+4)(s-4)}$$

• You need to apply partial decomposition

$$\frac{A}{s} + \frac{B}{(s-1)} + \frac{C}{(s+4)} + \frac{d}{(s-4)} = \frac{s^3 + 7s^2 - s - 1}{s(s-1)(s+4)(s-4)}$$

$$A(s-1)(s+4)(s-4) + B(s)(s+4)(s-4) + C(s)(s-1)(s-4) + d(s)(s-1)(s+4)$$

$$= s^3 + 7s^2 - s - 1$$

you can use the cover-up method on this to get what A, B, C and D are :

$$A = \frac{1}{16}, \quad B = -\frac{8}{15}, \quad C = -\frac{53}{160}, \quad D = \frac{173}{96}$$

$$\mathcal{L}\left\{ \frac{1}{16}\frac{1}{s} - \frac{8}{15}\frac{1}{(s-1)} - \frac{53}{160}\frac{1}{(s+4)} + \frac{173}{96}\frac{1}{(s-4)} \right\}$$

$$\boxed{y(t) = \frac{1}{16} - \frac{8}{15}e^{t} - \frac{53}{160}e^{-4t} + \frac{173}{96}e^{4t}}$$

we found "y" we now need to find "x" we can just plug everything into either one of the two given equations

$$y'(t) = -\frac{8}{15}e^{t} + \frac{53}{40}e^{-4t} + \frac{173}{24}e^{4t}$$

$$\frac{dy}{dt} = 8x - t$$

$$\left(-\frac{8}{15}e^{t} + \frac{53}{40}e^{-4t} + \frac{173}{24}e^{4t}\right) = 8x - t$$

$$-\frac{8}{15}e^{t} + \frac{53}{40}e^{-4t} + \frac{173}{24}e^{4t} + t = 8x$$

$$\boxed{x(t) = -\frac{1}{15}e^{t} + \frac{53}{320}e^{-4t} + \frac{173}{192}e^{4t} + \frac{1}{8}t}$$

using the laplace trans form to solve the given system of differential equation

$$\begin{cases} \dfrac{dx}{dt} = -x + y \\ \dfrac{dy}{dt} = 2x \end{cases} \qquad x(0) = 0, \quad y(0) = 5$$

$$\begin{cases} x' + x - y = 0 \\ y' - 2x = 0 \end{cases} \Longrightarrow \begin{cases} \mathcal{L}\{x'\} + \mathcal{L}\{x\} - \mathcal{L}\{y\} = 0 \\ \mathcal{L}\{y'\} - 2\mathcal{L}\{x\} = 0 \end{cases}$$

$$\begin{cases} sx(s) - x(0) + x(s) - y(s) = 0 \\ sy(s) - y(0) - 2x(s) = 0 \end{cases} \Longrightarrow \begin{cases} sx(s) + x(s) - y(s) = 0 \\ sy(s) - 2x(s) = 5 \end{cases}$$

if we multiply the first equation by s we can eliminate x

$$\begin{cases} s\,(sx(s) + x(s) - y(s) = 0) \\ sy(s) - 2x(s) = 5 \end{cases} \Longrightarrow \begin{cases} s^2x(s) + sx(s) - \cancel{sy(s)} = 0 \\ \cancel{sy(s)} - 2x(s) = 5 \end{cases}$$

$$s^2x(s) + sx(s) - 2x(s) = 5$$

$$x(s)\left[s^2 + s - 2\right] = 5$$

$$x(s)\,(s+2)(s-1) = 5$$

• you need to apply Partial decomposition

$$x(s) = \frac{5}{(s+2)(s-1)}$$

$$\left[\frac{A}{(s+2)} + \frac{B}{(s-1)} = \frac{5}{(s+2)(s-1)}\right] (s+2)(s-1)$$

$$A(s-1) + B(s+2) = 5$$

You can apply the cover up method to get the values for A and B

$$A = -\frac{5}{3}, \quad B = \frac{5}{3}$$

$$\left\{ -\frac{5}{3} \cdot \frac{1}{s+2} + \frac{5}{3} \frac{B}{s-1} \right\}$$

$$\boxed{x(t) = -\frac{5}{3} e^{-2t} + \frac{5}{3} e^{t}}$$

$$x'(t) = \frac{10}{3} e^{-2t} + \frac{5}{3} e^{t}$$

$$\frac{dx}{dt} = -x + y$$

$$\left(\frac{10}{3} e^{-2t} + \frac{5}{3} e^{t}\right) = -\left(-\frac{5}{3} e^{-2t} + \frac{5}{3} e^{t}\right) + y$$

$$\frac{10}{3}e^{-2t} + \frac{5}{3}e^{t} = \frac{5}{3}e^{-2t} - \frac{5}{3}e^{t} + y$$

$$\boxed{\frac{5}{3}e^{-2t} + \frac{10}{3}e^{t} = y(t)}$$

use the laplace transform to solve the given system of differential equations

$$\begin{cases} \dfrac{dx}{dt} = x - 2y \\[2ex] \dfrac{dy}{dt} = 5x - y \end{cases} \qquad x(0) = -1, \qquad y(0) = 5$$

$$\begin{cases} x' - x + 2y = 0 \\ y' - 5x + y = 0 \end{cases} \Longrightarrow \begin{cases} \mathcal{L}\{x'\} - \mathcal{L}\{x\} + 2\mathcal{L}\{y\} = 0 \\ \mathcal{L}\{y'\} - 5\mathcal{L}\{x\} + \mathcal{L}\{y\} = 0 \end{cases}$$

$$\begin{cases} sx(s) - x(0) - x(s) + 2y(s) = 0 \\ sy(s) - y(0) - 5x(s) + y(s) = 0 \end{cases} \Longrightarrow \begin{cases} sx(s) - (-1) - x(s) + 2y(s) = 0 \\ sy(s) - 5 - 5x(s) + y(s) = 0 \end{cases}$$

$$\begin{cases} sx(s) - x(s) + 2y(s) = -1 \\ sy(s) - 5x(s) + y(s) = 5 \end{cases} \Longrightarrow \begin{cases} x(s)(s-1) + 2y(s) = -1 \\ -5x(s) + sy(s) + y(s) = 5 \end{cases}$$

if we multiply the first equation with "s" and multiply the second equation by "(s-1)" we can cancel out x's and that would leave us with y's

$$\begin{cases} s\left(x(s)\,(s-1) + 2y(s) = -1\right) \\ (s-1)\left(-5x(s) + y(s)\,(s+1) = s\right) \end{cases} \Longrightarrow \begin{cases} \cancel{5x(s)\,(s-1)} + 10y(s) = -s \\ \cancel{-5x(s)(s-1)} + y(s)(s^2-1) = s(s-1) \end{cases}$$

$$10y(s) + y(s)\,(s^2-1) = -s + s^2 - s$$

$$y(s)\left[10 + s^2 - 1\right] = s^2 - 2s$$

$$y(s)\left[s^2 + 9\right] = s^2 - 2s$$

$$y(s) = \frac{s^2 - 2s}{s^2 + 9}$$

$$y(s) = \mathcal{L}^{-1}\left\{ s\left(\frac{s}{s^2+9}\right) - \frac{2}{3}\cdot\frac{3}{s^2+9} \right\}$$

$$\boxed{y(t) = s\cos(3t) - \frac{2}{3}\sin(3t)}$$

$$y'(t) = -3s\sin(3t) - 2\cos(3t)$$

$$\frac{dy}{dt} = 5x - y$$

$$\left(-3s\sin(3t) - 2\cos(3t)\right) = 5x - \left(s\cos(3t) - \frac{2}{3}\sin(3t)\right)$$

$$-15\sin(3t) - 10\cos(3t) = 5x - 5\cos(3t) + \frac{10}{3}\sin(3t)$$

$$5x = -5\cos(3t) - \frac{55}{3}\sin(3t)$$

$$\boxed{x = -\cos(3t) - \frac{11}{3}\sin(3t)}$$

if you need this all need to do is take inverse: $1 = \mathscr{L}^{-1}\left\{\frac{1}{s}\right\}$

Theorem 7.1.1 **Transforms of Some Basic Functions**

(a) $\mathscr{L}\{1\} = \frac{1}{s}$

(b) $\mathscr{L}\{t^n\} = \frac{n!}{s^{n+1}}, n = 1, 2, 3, \ldots$

(c) $\mathscr{L}\{e^{at}\} = \frac{1}{s-a}$

(d) $\mathscr{L}\{\sin kt\} = \frac{k}{s^2+k^2}$

(e) $\mathscr{L}\{\cos kt\} = \frac{s}{s^2+k^2}$

(f) $\mathscr{L}\{\sinh kt\} = \frac{k}{s^2-k^2}$

(g) $\mathscr{L}\{\cosh kt\} = \frac{s}{s^2-k^2}$